Ayaz Khan

New York, NY | U.S. Citizen | +1 (917) 943-3853 | aak2259@columbia.edu
linkedin.com/in/ayazkhan27 | github.com/ayazkhan27 | ORCID: 0009-0008-5695-4934

Education

Columbia University New York, NY
Master of Arts in Statistics, Graduate School of Arts and Sciences *Expected Dec 2026*

- **Honors:** National Science Foundation Graduate Research Fellow, 1 of 1,000 awardees from 17,000 applicants (top 5.9%), $159,000 research award.
- **Coursework:** Statistical Machine Learning, Probability, Statistical Inference, Linear Regression Models, Elementary Stochastic Processes.

CUNY Hunter College New York, NY
Special Honors B.A. in Statistics, Minor in Computer Science, cum laude *Dec 2024*

Professional Experience

Telenav Santa Clara, CA
Applied Machine Learning and Data Engineering Intern *Jun 2026 – Aug 2026*

- Fine-tuned a 270M-parameter FunctionGemma router in PyTorch to convert analyst questions into schema-valid calls across eight typed read-only tools, reaching **95.5% function accuracy**, **86.4% exact match** on 500 held-out prompts, and **95.4% serving accuracy** against live Snowflake.
- Established a nonparametric accuracy ceiling of **56.69%** (95% CI [55.13, 58.08]) on 4,338 true-ping rows, proving the production calibrator ran **9 points below the bound** and cancelling a release that would have degraded pricing.
- Piloted privileged features distillation with a quote-stage teacher, lifting ping-stage test AUC from **0.6206 to 0.6457 (+2.5 points)** across 23,559 holdout rows with no change to serve-time features.
- Shipped an analyst explainability surface routing questions to deterministic Snowflake SQL, using LoRA and full fine-tuning, GGUF Q8_0 quantization, and grammar-constrained decoding behind a 25/25 faithfulness gate.
- Diagnosed target leakage and non-temporal splits in an acquisition backtest, rebuilding it with user-grouped temporal validation to correct an inflated 0.806 AUC to an honest **0.7401** and surface **$127K** of campaign waste.

Calx New York, NY
Founder and Lead Researcher, Cross-Domain Scientific Discovery Engine *2025 – Present*

- Architected a neuro-symbolic discovery engine porting the Structure-Mapping Engine to force cross-domain analogical reasoning over published research; filed **U.S. Provisional Patent 64/029,090** through Columbia Technology Ventures.
- Generated two novel square-root Kalman filter variants with no identified prior art, and a Certified Affine Corridor dynamic program achieving a **31.55% mean reduction in operations at 45/45 exact correctness**.
- Published **SMA-1** (`structuremappingmemory`) to PyPI, a structure-mapping retrieval layer over 11 ontologies and 594,000 concepts shipping as an MCP server, beating the strongest RAG baseline on long-tail top-5 retrieval by **+0.34** (medicine) and **+0.17** (genomics); deployed with Docker and CI/CD on Google Cloud.

Data Science Institute, Columbia University New York, NY
DSI Scholar, NYSERDA Energy Insecurity and Inequities Research *Jan 2026 – May 2026*

- Built an NLP pipeline over **317 Con Edison rate-case testimony PDFs** filed with the NY Public Service Commission, extracting **12,073 segments** with pdfplumber and routing exhibits and procedural artifacts away from **10,225 substantive segments**.
- Modeled the corpus with LegalBERT embeddings, UMAP, and HDBSCAN via BERTopic, consolidating 22 clusters into **16 docket themes** and showing affordability testimony fell **3.8 percentage points** from 2022 to 2025 while rate-design and IT-systems testimony rose **4.9 and 3.4**.

National Science Foundation Kingsville, TX
Research Experiences for Undergraduates (REU) Researcher *Jun 2024 – Aug 2024*

- Created the Intrusion Effort Index, a novel wireless sensor network security metric derived from aperiodic tiling, and processed sensor data in Python and NumPy for a **15% efficiency improvement** in 3D deployments; published **first author at IEEE ICNGN 2024**.

Academic and Research Projects

Credit Card Fraud Detection Pipeline Python, XGBoost, Dash
Bayesian Networks, scikit-learn, SMOTE-NC *Jan 2025*

- Engineered a detection system for a severely imbalanced **0.172% positive-rate** dataset using SMOTE-NC resampling, Bayesian networks for causal inference, and probability-calibrated XGBoost for cost-asymmetric ranking; deployed a monitoring dashboard in Dash.

Technical Skills

- **Programming:** Python, R, SQL, Java, C++, SAS.
- **Machine Learning:** PyTorch, scikit-learn, transformers, BERTopic, UMAP, HDBSCAN, XGBoost, LightGBM, LoRA.
- **Data and Infrastructure:** pandas, NumPy, Snowflake, pdfplumber, Docker, Git, CI/CD, Google Cloud, AWS EC2, Linux.
- **Analytics:** Tableau, Bloomberg Terminal, Dash, matplotlib, ggplot2, MATLAB, Jupyter, LaTeX.
- **Methods:** Causal Inference, Bayesian Networks, Stochastic Processes, Survival Analysis, SHAP, A/B Testing.